

1. IMPLEMENTATION OF STACK USING LINKED LIST

```
#include<stdio.h>

#include<stdlib.h>

struct Node
{
    int data;
    struct Node *next;
};

struct Node *head = NULL;

void push(int val)
{
    //create new node
    struct Node *newNode = malloc(sizeof(struct Node));
    newNode->data = val;

    //make the new node points to the head node
    newNode->next = head;

    //make the new node as head node
    //so that head will always point the last inserted data
    head = newNode;
}

void pop()
{
    //temp is used to free the head node
    struct Node *temp;
```

```
if(head == NULL)
    printf("Stack is Empty\n");
else
{
    printf("Poped element = %d\n", head->data);

    //backup the head node
    temp = head;

    //make the head node points to the next node.
    //logically removing the node
    head = head->next;

    //free the popped element's memory
    free(temp);
}

//print the linked list
void display()
{
    struct Node *temp = head;

    //iterate the entire linked list and print the data
    while(temp != NULL)
    {
        printf("%d->", temp->data);
        temp = temp->next;
    }
    printf("NULL\n");
}
```

```

}

int main()
{
    push(70);
    push(80);
    push(90);
    push(100);
    printf("Linked List\n");
    display();
    pop();
    printf("After the pop, the new linked list\n");
    display();
    pop();
    printf("After the pop, the new linked list\n");
    display();

    return 0;
}

```

Output:

Linked List

100->90->80->70->NULL

Poped element = 100

After the pop, the new linked list

90->80->70->NULL

Poped element = 90

After the pop, the new linked list

80->70->NULL

2. IMPLEMENTATION OF QUEUE USING LINKED LIST

```
#include <stdio.h>
#include <stdlib.h>

struct node
{
    int data;
    struct node* next;
}; //defining linked list to implement queue

struct node *front = NULL;
struct node *rear = NULL;

void enqueue(int d) //function to insert a node in queue
{
    struct node* new_n;
    new_n = (struct node*)malloc(sizeof(struct node));
    new_n->data = d;
    new_n->next = NULL;
    if((front == NULL)&&(rear == NULL)){
        front = rear = new_n;
    }
    else{
        rear->next = new_n;
        rear = new_n;
    }
}

void display() //function to display the queue
{
    struct node* temp;
    if((front == NULL)&&(rear == NULL)){
        printf("\nQueue is Empty");
    }
    else{
        temp = front;
        while(temp){
            printf(" %d ", temp->data);
            temp = temp->next;
        }
    }
}

void dequeue() //function to delete an element from a queue
{
    struct node *temp;
    temp = front;
```

```

        if((front == NULL)&&(rear == NULL)){
            printf("\nQueue is Empty");
        }
        else{
            front = front->next;
            free(temp);
        }
    }
}

int main();//main function to use all our declared function
{
    enqueue(5);
    enqueue(10);
    enqueue(15);
    enqueue(20);
    enqueue(25);
    printf("Queue:");
    display();
    printf("\nQueue After Dequeue:");
    dequeue();
    display();
}

```

OUTPUT

Queue: 5 10 15 20 25

Queue After Dequeue: 10 15 20 25